

PROPOSAL ABSTRACT

Name of Principal Investigator:	Manuel Tironi
Proposal Title:	Wild boar socioecologies: invasive species as contact zones in the peweñantu, Southern Los Andes

- Provide a general overview of the key elements to be addressed in the proposal, such as the problem statement, objectives, methodology, and expected outcomes.

Invasive alien species (IAS) have emerged as a crucial challenge for the livelihood of diverse socioecological communities. The singularity of the threat, as shown by the international literature, is that it produces ecological, economic, social, and cultural impacts that cannot be disentangled into discrete factors but require an integrated and synergistic approach. Moreover, local communities are demanding to be recognized as valid knowledge producers, with the capacity to articulate robust evidence and hypotheses for the management of IAS —evidence and hypotheses that, however, have been seldom validated by scientific research.

The aim of this project is **to articulate a novel program for a transdisciplinary approach to invasive alien species (IAS)**. Our leading hypothesis is that, rather than discrete entities that produce multiple yet separated effects —whether biological, social, ecological, or political— IAS are better understood as **contact zones** (Haraway 2003), that is, as nodes of relation between entities, knowledges, epistemological systems, and actors in constant and reciprocal exchange. To this end, the project grounds its questions in the case of **wild boars in the Andean Araucanía**, an unparalleled case to understand the tensions around *Sus scrofa* for its world-renowned biodiversity and intercultural richness. Taking an explicit transdisciplinary approach, the project sits at the intersection between conservation ecology, cultural anthropology, and science and technology studies, but also —and crucial for its pluri-epistemological take— formally integrating **Mapuche kimün (knowledge)** in the entire research cycle, an epistemological breakthrough in the Chilean context.

The main question of the project is **how to analyze and understand the socioecological impacts of *Sus scrofa* in the Andean Araucanía from a transdisciplinary, intercultural, and multispecies perspective, engaging in the proposal of community-based strategies for the management of invasive species**. This question will be tackled through 4 objectives: **Obj. 1**, to characterize and understand the ecology of *Sus scrofa* (population size, mobility patterns, and diet); **Obj. 2**, to describe and analyze the history and everydayness of co-habitation with wild boars in the context of Indigenous and campesino life projects in the Andean Araucanía; **Obj. 3**, to register and systematize local knowledge systems about —and practices of— tracking, hunting, and managing *Sus scrofa*; and **Obj. 4**, to integrate research results around the proposition of community-based programs and interventions for multi-actor management of invasive species.

Methodologically, the project follows an “intra-action model” in which we combine eco-biological and socio-anthropological methods in a process of cross-fertilization: biological, anthropological, ecological, sociological, and Mapuche insights will be integrated —thus blurring its borders— *within* each objective, avoiding the usual assignation of one objective for each method. Concretely, each objective is articulated around a combination of **Field ecology methods** (camera traps; population models; line transect surveys; mobility and distribution analysis with drone thermal cameras; dietary analysis; and landscape analyses); **Ethnographic methods** (in-depth interviews; participant-observation fieldwork; and site visits to communities and campesino families); and **Participatory methods** (community mapping exercises, co-design workshops, and *trawun* [assemblies]).

The project's main case is Villarrica National Park's Puesco-Lanin sector based on a groundbreaking collaboration with the Winkulmapu Indigenous Association (WIA) constituted by Mapuche communities from Curarrehue. WIA signed a Governance Agreement with the National Forestry Corporation (CONAF) and the Ministry of National Assets to manage 17,000-hectare subsection of the Puesco-Lanin sector. This is **an unprecedented collaboration for the advancement of transdisciplinary, intercultural, and collaborative socioecological research on IAS**, a unique (co)laboratory not only for developing a research program on the complexities of AIS as contact zones in the Andean Araucanía, but also to transfer scalable and actionable guidelines for Chile's new conservation policies (SBAP).

We aim at producing **top-quality scientific papers**, but also at integrating the results from each objective in the formulation of **concrete, community-oriented, and co-designed programs for *Sus scrofa* management**, as well as in several events and outputs for transdisciplinary exchange: we aim at producing **a community book, policy briefs, and an art exhibition with local artists** to generate an open conversation on the territorial challenges of wild boars.

The team's research experience conveys the needed expertise for the project. Their prior and ongoing collaborations —added to the fact that all of them live and work in the Araucanía region— secure a robust workflow. And crucially, a **Mapuche kintukafe (researcher) will formally participate in the project representing WIA in all four objectives, thus articulating an unprecedented pluri-epistemological, transdisciplinary research architecture**.